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2. Please introduce the Divide-and-Conquer algorithm and write out its general steps used to solve problem? (10 marks)

Score:

Answer:

1. The divide-and-conquer strategy is a powerful paradigm for designing efficient algorithms. This approach first divides a problem into two smaller sub-problems and each sub-problem is identical to its original problem, Both of them are then solved and the sub-solutions are finally merged into the final solution.

(5points)

2. The general steps of divide-and-conquer strategy used to solve problem are:

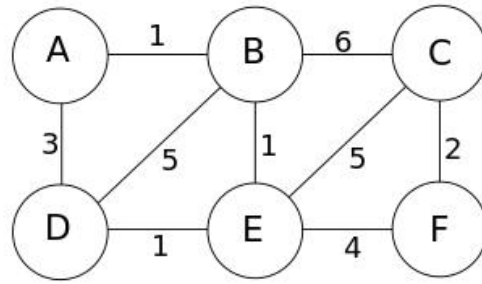
(5points)

Step 1: If the problem size is small, solve this problem directly; otherwise, split the original problem into 2 sub-problems with equal sizes.

Step 2: Recursively solve these 2 sub-problems by applying this algorithm.

Step 3: Merge the solutions of the 2 sub-problems into a solution of the original problem.

3. Given an un-directed graph like below, please find out its minimum spanning trees (MST) using Kruskal's and Prim's algorithm respectively. (15 marks)



Score:

Answer:

1. To use Kruskal algorithm solving it as below: **(5points)**

Step 1 To sort all of edges by increasing order like below,

AB(1), BE(1), DE(1), CF(2), AD(3), EF(4), BD(5), CE(5), BC(6)

Step 2 To make the minimum spanning trees (MST) with selecting the edges one by one from the up sorted queue

(1) AB(1)

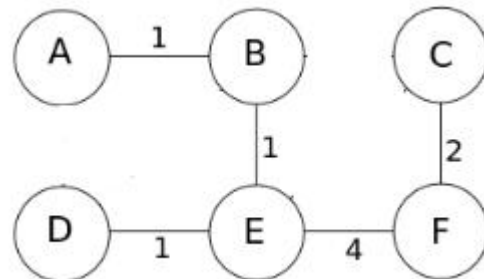
(2) BE(1)

(3) DE(1)

(4) CF(2)

(5) AD(3) Rejected with cycle

(6) EF(4)



The MST

Step 3 the minimum cost for this MST is: $1+1+1+2+4=9$

2. To use Prim algorithm solving it as below: **(10points)**

Step 1 To sort all of edges by increasing order like below,

AB(1), BE(1), DE(1), CF(2), AD(3), EF(4), BD(5), CE(5), BC(6), and then select the first edge AB into MST and the A and B set change like below:

Step 2 There are the edges could be selected at this step and sort them BE(1), AD(3), BD(5), BC(6), and

A	B = V-A	
A, B	C, D, E, F	Step 1
A, B, E	C, D, F	Step 2
A, B, E, D	C, F	Step 3
A, B, E, D, F	C	Step 4
A, B, E, D, F, C	Φ	Step 5

then the edge BE(1) will be selected at this step, and the sets A and B will be changing like above.

Step 3 there are these edges will be selected and sorting them before like this: ED(1), AD(3), EF(4), BD(5), CE(5), BC(6), and the edge ED(1) will be selected at this step, and the sets A and B will be changed like above.

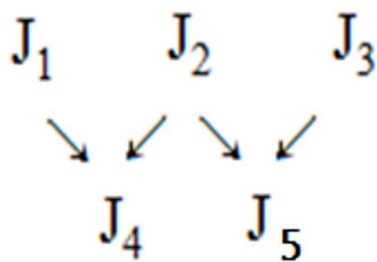
Step 4 there are these edges will be selected and sorting them before like this: EF(4), CE(5), BC(6), and the edge EF will be selected at this step and the sets A and B will be changing like above.

Step 5 there are these edges will be selected and sorting them before like this: CF(2), CE(5), BC(6), and the edge CF will be selected into MST at this step and the sets A and B will changing like above.

The result likes the result of Kruskal's algorithm .

4. There are five jobs needed to be assigned to five persons. Given us the following job assignment condition and the cost matrix, please write out the solution tree and get all possible solutions, in the meantime to calculate the reduced cost matrix, according to which you get the lower bound and the optimal solution using tree searching algorithm. (20 marks)

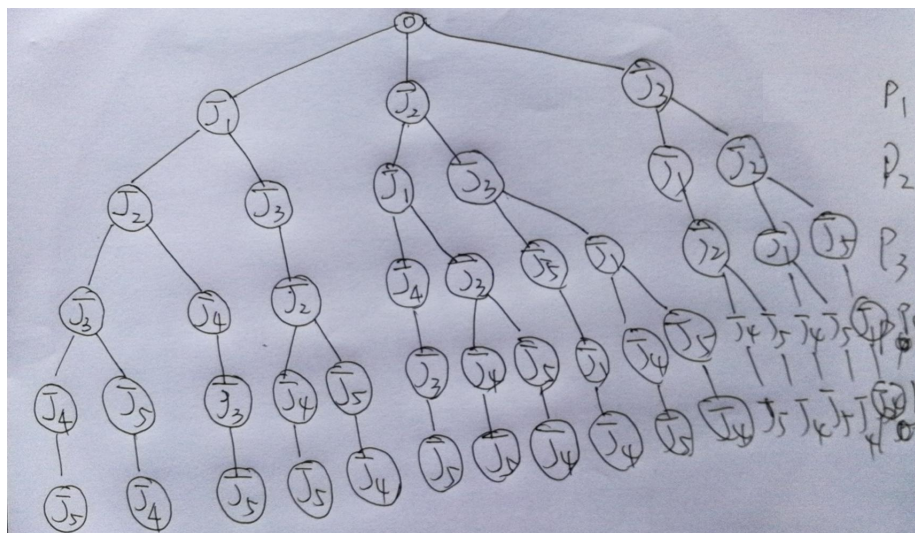
Score:



Jobs Persons	1	2	3	4	5
1	29	19	17	12	10
2	32	30	26	28	22
3	3	21	7	9	8
4	18	13	10	15	16
5	6	23	12	20	33

Answer:

1. the solution tree likes the graph below.



(3points)

And all possible solutions are: J1J2J3J4J5, J1J2J4J3J5, J1J3J2J4J5, J1J3J2J5J4, J2J1J4J3J5, J2J1J3J4J5, J2J1J3J5J4,

J2J3J5J1J4, J2J3J1J4J5, J2J3J1J5J4, J3J1J2J4J5, J3J1J2J5J4, J3J2J5J1J4 with total 16 possible solutions. **(2points)**

2. the reduced cost matrix is:

Jobs Persons	1	2	3	4	5	
1	19	6	7	0	0	10
2	12	5	4	4	0	22
3	0	15	4	4	5	3
4	8	0	0	3	6	10
5	0	14	6	12	27	6
		3		2		

With the lower bound:

$$10+22+3+10+6+2+3 = 56$$

3. The optimal solution using tree searching algorithm like:

1) Root -> J1 with cost (19 +

56)=75, Root -> J2 with cost (6 + 56)=62, and Root -> J3 with cost (7 + 56)=63; **(5points)**

(8points) searching process

2) Then expand Root -> J2: Root -> J2->J1 with cost (6 + 56) + 12 =74, and Root -> J2-> J3 with cost (6 + 56) + 4 =66;

3) and Then expand Root -> J3 with cost 63: Root -> J3 -> J1 with cost 63+12 =75, and Root -> J3 -> J2 with cost 63+5=68;

4) Then expand Root -> J2-> J3 with cost 66: Root -> J2-> J3->J1 with cost 66 +0 =66, and Root -> J2-> J3->J5 with cost 66 +5 =71;

5) Then expand Root -> J2-> J3 ->J1 with cost 66: Root -> J2-> J3 ->J1->J4 with cost 66 +3 =69, and Root -> J2-> J3 ->J1->J5 with cost 66 +6 =72;

6) Then expand Root -> J3 -> J2 with 68: Root -> J3 -> J2->J1

- with $68+0=68$, and Root $\rightarrow$ J3 $\rightarrow$ J2 $\rightarrow$ J5 with $68+5=73$;
- 7) Then expand Root $\rightarrow$ J3 $\rightarrow$ J2 $\rightarrow$ J1 with 68: Root $\rightarrow$ J3 $\rightarrow$ J2 $\rightarrow$ J1 $\rightarrow$ J4 with $68+3=71$, and Root $\rightarrow$ J3 $\rightarrow$ J2 $\rightarrow$ J1 $\rightarrow$ J5 with $68+6=74$;
- 8) Then expand Root $\rightarrow$ J2 $\rightarrow$ J3 $\rightarrow$ J1 $\rightarrow$ J4 with cost 69: Root $\rightarrow$ J2 $\rightarrow$ J3 $\rightarrow$ J1 $\rightarrow$ J4 $\rightarrow$ J5 with cost $69+27=96$; (leaf)
- 9) Then expand Root $\rightarrow$ J3 $\rightarrow$ J2 $\rightarrow$ J1 $\rightarrow$ J4 with 71: Root $\rightarrow$ J3 $\rightarrow$ J2 $\rightarrow$ J1 $\rightarrow$ J4 $\rightarrow$ J5 with cost $71+27=98$; (leaf)
- 10) Then expand Root $\rightarrow$ J2 $\rightarrow$ J3 $\rightarrow$ J1 $\rightarrow$ J5 with 72: Root $\rightarrow$ J2 $\rightarrow$ J3 $\rightarrow$ J1 $\rightarrow$ J5 $\rightarrow$ J4 with cost $72+12=84$; (leaf)
- 11) Then expand Root $\rightarrow$ J3 $\rightarrow$ J2 $\rightarrow$ J5 with 73: Root $\rightarrow$ J3 $\rightarrow$ J2 $\rightarrow$ J5 $\rightarrow$ J1 with cost $73+8=81$;
- 12) Then expand Root $\rightarrow$ J3 $\rightarrow$ J2 $\rightarrow$ J1 $\rightarrow$ J5 with 74: Root $\rightarrow$ J3 $\rightarrow$ J2 $\rightarrow$ J1 $\rightarrow$ J5 $\rightarrow$ J4 with cost: $74++12=86$; (leaf)
- 13) Then expand Root $\rightarrow$ J3 $\rightarrow$ J1 with cost 75: Root $\rightarrow$ J3 $\rightarrow$ J1 $\rightarrow$ J2 with cost: $75+15=90$;
- 14) Then expand Root $\rightarrow$ J1 with cost 75: Root $\rightarrow$ J1 $\rightarrow$ J2 with cost $75+5=80$, and Root $\rightarrow$ J1 $\rightarrow$ J3 with cost $75+4=79$;
- 15) Then expand Root $\rightarrow$ J1 $\rightarrow$ J3 with cost 79: Root $\rightarrow$ J1 $\rightarrow$ J3 $\rightarrow$ J2 with cost $79+15=94$;
- 16) Then expand Root $\rightarrow$ J1 $\rightarrow$ J2 with cost 80: Root $\rightarrow$

J1->J2->J3 with cost $80+4=84$, and Root -> J1->J2->J4 with cost $80+4=84$;

17) Then expand Root -> J3 -> J2->J5->J1 with cost 81: Root -> J3 -> J2->J5->J1-J4 with cost: $81+12=93$; (leaf)

18) the optimal solution is Root -> J2-> J3 ->J1->J5->J4 with cost $72+12=84$; (leaf); Or J2 -P1, J3-P2, J1-P3, J5-P4 and J4-P5 with minimum cost 84. **(2points) result**

5. Given you five matrices: $M_1(10 \times 20)$, $M_2(20 \times 30)$, $M_3(30 \times 20)$, $M_4(20 \times 10)$, $M_5(10 \times 20)$, please calculate the minimum cost of their product using dynamic programming algorithm for matrices multiplication and write out the optimal multiplication sequence with minimum cost. (20 marks)

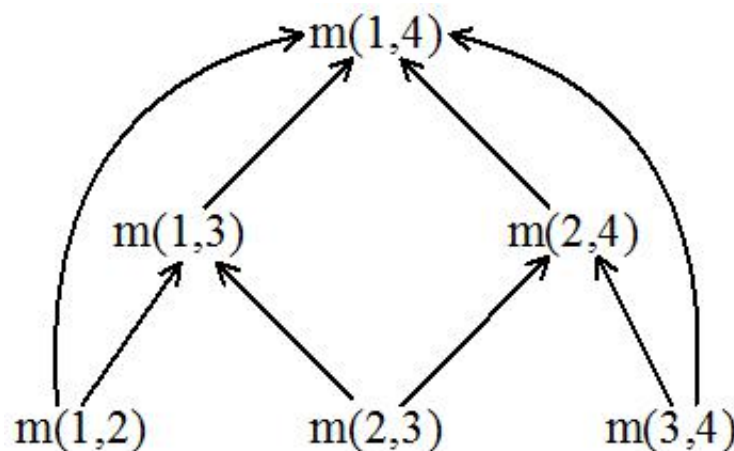
Answer:

Score:

1. Let $m(i, j)$ denote the minimum cost for computing $A_i A_{i+1} \cdots A_j$

$$m(i, j) = \begin{cases} 0 & \text{if } i = j \\ \min_{i \leq k \leq j-1} \{m(i, k) + m(k+1, j) + p_{i-1}p_kp_j\} & \text{if } i < j \end{cases}$$

Computation sequence



(3points)

2. For $M_1(10 \times 20)$, $M_2(20 \times 30)$, $M_3(30 \times 20)$, $M_4(20 \times 10)$, $M_5(10 \times 20)$, we calculate them in the following half table.

1) The members in the first row calculated: $M_{12} = 10 \times 20 \times 30$

=6000 with new matrix size: 10×30 ;

M11=0	M22=0	M33=0	M44=0	M55=0
M12=6000	M23=12000	M34=6000	M45=4000	
M13=12000	M24=12000	M35=12000		<u>(3points)</u>
M14=14000	M25=16000			
M15=16000				

$M23 = 20 \times 30 \times 20 = 12000$ with new matrix size: 20×20 ;

$M34 = 30 \times 20 \times 10 = 6000$ with new matrix size: 30×10 ;

$M45 = 20 \times 10 \times 20 = 4000$ with new matrix size: 20×20 ; **(3points)**

2) The members in the second row calculated:

(1) $M13 = (M12 \times M3)$ or $(M1 \times M23)$

For $(M12 \times M3)$, it is: $6000 + 10 \times 30 \times 20 = 12000$ with new size: 10×20 ;

For $(M1 \times M23)$, it is: $12000 + 10 \times 20 \times 20 = 16000$ with new size: 10×20 ;

The minimum value is : 12000. **(3points)**

(2) $M24 = (M23 \times M4)$ or $(M2 \times M34)$

For $(M23 \times M4)$, it is: $12000 + 20 \times 20 \times 10 = 16000$ with new size: 20×10 ;

For $(M2 \times M34)$, it is: $6000 + 20 \times 30 \times 10 = 12000$ with new size: 20×10 ;

The minimum value is : 12000;

$$(3) \ M_{35} = (M_{34} \times M_5) \text{ or } (M_3 \times M_{45})$$

For $(M_{34} \times M_5)$, it is: $6000 + 30 \times 10 \times 20 = 12000$ with new size: 30×20 ;

For $(M_3 \times M_{45})$, it is: $4000 + 30 \times 20 \times 20 = 16000$ with new size: 30×20 ;

The minimum value is : 12000;

3) The members in the third row calculated: **(3points)**

$$(1) \ M_{14} = (M_{13} \times M_4) \text{ or } (M_{12} \times M_{34}) \text{ or } (M_1 \times M_{24})$$

For $(M_{13} \times M_4)$, it is: $12000 + 10 \times 20 \times 10 = 14000$ with new size: 10×10 ;

For $(M_{12} \times M_{34})$, it is: $6000 + 6000 + 10 \times 30 \times 10 = 15000$ with new size: 10×10 ;

For $(M_1 \times M_{24})$, it is: $12000 + 10 \times 20 \times 10 = 14000$ with new size: 10×10 ;

The minimum value is : 14000;

$$(2) \ M_{25} = (M_{24} \times M_5) \text{ or } (M_{23} \times M_{45}) \text{ or } (M_2 \times M_{35})$$

For $(M_{24} \times M_5)$, it is: $12000 + 20 \times 10 \times 20 = 16000$ with new size: 20×20 ;

For $(M_{23} \times M_{45})$, it is: $12000 + 4000 + 20 \times 20 \times 20 = 24000$ with new size: 20×20 ;

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For $(M2 \times M35)$, it is: $12000 + 20*30*20 = 24000$ with new size: 20×20 ;

The minimum value is : 16000;

4) The members in the last row calculated: **(3points)**

$M15 = (M1 \times M25) \text{ or } (M12 \times M35) \text{ or } (M13 \times M45) \text{ or } (M14 \times M5)$

For $(M1 \times M25)$, it is: $16000 + 10*20*20 = 20000$ with new size: 10×20 ;

For $(M12 \times M35)$, it is: $6000 + 12000 + 10*30*20 = 24000$ with new size: 10×20 ;

For $(M13 \times M45)$, it is: $12000 + 4000 + 10*20*20 = 20000$ with new size: 10×20 ;

For $(M14 \times M5)$, it is: $14000 + 10*10*20 = 16000$ with new size: 10×20 ;

The minimum value for $M15$ is : 16000, this is the minimum cost of their product, and the optimal multiplication sequence is: **(2points)**

$(M1 \times (M2 \times (M3 \times M4))) \times M5$

or

$((M1 \times M2) \times M3) \times M4 \times M5$

6. Given us four objects which would be put into a knapsack with capacity of 16, and their weights and profits are shown at the

i	W_i	P_i
1	10	40
2	5	25
3	6	35
4	8	45

$M=20$

following table, please calculate its optimal solution for knapsack problem using greedy algorithm and for 0/1 knapsack problem using dynamic programming algorithm independently.

(15 marks)

Score:

Answer:

The capacity of this knapsack could be 16 or 20 with no problem. We solve it by the capacity 20 like below:

1. For knapsack problem using greedy algorithm, to calculate the profit per weight like: **(5 points)**

1) $P_1/W_1 = 40/10 = 4$, $P_2/W_2 = 25/5 = 5$, $P_3/W_3 = 35/6 = 5.833$,
 $P_4/W_4 = 45/8 = 5.625$

2) To sort them by decrease order:

$P_3/W_3 > P_4/W_4 > P_2/W_2 > P_1/W_1$

3) To select the item one by one using the greedy algorithm:

Step 1 (W_3 , P_3) = (6, 35) to be selected, then
 $M' = (M - W_3) = 20 - 6 = 14$;

Step 2 (W_4 , P_4) = (8, 45) to be selected, then

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$$M''=(M'-W4)=14-8=6;$$

Step 3, $(W2, P2) = (5, 25)$ to be selected, then
 $M'''=(M''-W2)=6-5=1;$

Step 4, one of $(W1, P1) = (1, 4)$ to be selected, then
 $M''''=(M'''-W1/10)=1-10/10=0;$

Then the optimal solution for knapsack problem using greedy algorithm is: $(W3, P3) = (6, 35)$, $(W4, P4) = (8, 45)$, $(W2, P2) = (5, 25)$ and one of $(W1, P1) = (1, 4)$ are selected into this knapsack with maximum profits 109 and weight of 20.

If the capacity is 16, it is could solved like this:

$$4) P1/W1 = 40/10 = 4, P2/W2 = 25/5 = 5, P3/W3 = 35/6 = 5.833, \\ P4/W4 = 45/8 = 5.625$$

5) To sort them by decrease order:

$$P3/W3 > P4/W4 > P2/W2 > P1/W1$$

6) To select the item one by one using the greedy algorithm:

Step 1 $(W3, P3) = (6, 35)$ to be selected, then
 $M'=(M-W3)=16-6=10;$

Step 2 $(W4, P4) = (8, 45)$ to be selected, then
 $M''=(M'-W4)=10-8=2;$

Step 3, two of $(W2, P2) = 2*(1, 5)$ to be selected, then
 $M'''=(M''-W2)=2-2=0;$

Then the optimal solution for knapsack problem using greedy algorithm is: $(W_3, P_3) = (6, 35)$, $(W_4, P_4) = (8, 45)$, two of $(W_2, P_2) = (2, 10)$ are selected into this knapsack with maximum profits 90 and weight of 16.

2. For 0/1 knapsack problem using dynamic programming algorithm. **(10 points)**

1) Let $f_i(Q)$ be the value of an optimal solution to objects $1, 2, 3, \dots, i$ with capacity Q . then $f_i(Q) = \max\{f_{i-1}(Q), f_{i-1}(Q-W_i)+P_i\}$, the optimal solution of this problem is $f_4(20)$ or $f_4(16)$. **(1 points)**

2) To Calculate $f_4(20)$ like below:

$$(1) f_4(20) = \max\{f_3(20), f_3(20-8)+45\} = \max\{f_3(20), f_3(12)+45\} \quad \textbf{(1 points)}$$

$$(2) F_3(20) = \max\{f_2(20), f_2(20-6)+35\} = \max\{f_2(20), f_2(14)+35\}$$

$$f_3(12) = \max\{f_2(12), f_2(12-6)+35\} = \max\{f_2(12), f_2(6)+35\} \quad \textbf{(1 points)}$$

$$(3) f_2(20) = \max\{f_1(20), f_1(20-5)+25\} = \max\{f_1(20), f_1(15)+25\}$$

$$f_2(14) = \max\{f_1(14), f_1(14-5)+25\} = \max\{f_1(14),$$

$$f_1(9)+25\}$$

$$f_2(12) = \text{Max}\{f_1(12), f_1(12-5)+25\} = \text{Max}\{f_1(12), f_1(7)+25\}$$

$$f_2(6) = \text{Max}\{f_1(6), f_1(6-5)+25\} = \text{Max}\{f_1(6), f_1(1)+25\}$$

(2 points)

$$f_1(20) = 40 \ (20 > 10), f_1(15) = 40 \ (20 > 10), f_1(14) = 40 \ (20 > 10), f_1(9) = 0 \ (9 < 10), f_1(12) = 40 \ (20 > 10), f_1(7) = 0 \ (7 < 10), f_1(6) = 0 \ (6 < 10), f_1(1) = 0 \ (1 < 10);$$

(2 points)

(4) Replace them into the above (3)\(2) and (1), to get:

$$f_2(6) = 25,$$

$$f_2(12) = 25,$$

$$f_2(14) = \max\{40, 25\}=40,$$

$$f_2(20)=\text{Max}\{f_1(20), f_1(15)+25\} =\max\{40, 40+25\}=65;$$

Then

$$F_3(20)=\text{Max}\{f_2(20), f_2(14)+35\} =\max\{65, 40+35\}=75$$

$$F_3(12)=\text{Max}\{f_2(12), f_2(6)+35\}=\max\{25,25+35\}=60$$

(2 points)

In the end, we can get:

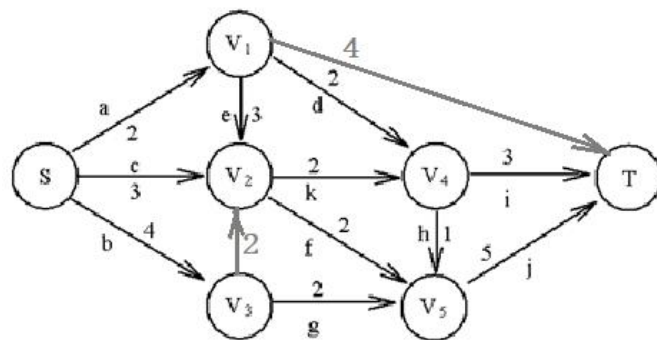
$$(5) \quad f_4(20)=\text{Max}\{f_3(20), f_3(12)+45\}=\max\{75, 60+45\}=105$$

The items $((W_4, P_4) = (8, 45), (W_3, P_3) = (6, 35), (W_2,$

P2) =(5, 25)) are selected into this knapsack with sum of weight $8+6+5=19 < 20$ and sum of profit $105 = 45+35+25$. **(1 points)**

3) To Calculate $f_4(16)$ almost like calculating $f_4(20)$.

7. Given us the following directed graph please find the shortest path from Node S to T using the A* algorithm.



Score:

Answer:

1) To define the cost function of node n : $f(n)$

$$f(n) = g(n) + h(n)$$

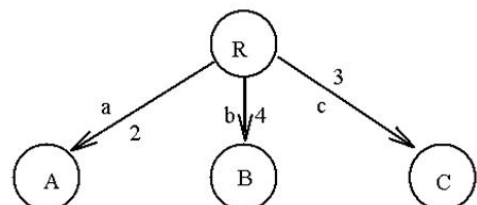
$g(n)$: cost from root to node n.

$h(n)$: estimated cost from node n to a goal node. **(2 points)**

2) To find the shortest path from Node S to T using the A* algorithm like below:

$$\text{Step 1: } f(A) = g(A) + h(A) = 2 + \min(2, 3, 4) = 2 + 2 = 4$$

$$f(B) = g(B) + h(B) = 4 + \min(2, 2) = 4 + 2 = 6$$



$$f(C) = g(C) + h(C) = 3 + \min(2, 2) = 3 + 2 = 5 \quad \text{(2 points)}$$

Step 2: Expand A node, then we can get:

$$f(D) = g(D) +$$

$$h(D) = f(A) + 2 + \min(1, 3) = 4 + 2 + 1 = 7$$

$$f(F) = g(F) + h(F) = f(A) + 4 + 0 = 2 + 4 = 6 \text{ (Destination)}$$

$$f(C) = g(C) + h(C) = 4 + 3 + \min(2, 2) = 9, \text{ and compare with step 1, } f(C) = 5; \quad \text{(2 points)}$$

Step 3: Expand C node, then we can get:

$$f(D) = g(D) + h(D) = f(C) + 2 + \min(3) = 5 + 2 + 3$$

$$= 10, \text{ compare with step 2, } f(D) = 7;$$

$$f(E) = g(E) + h(E) = f(C) + 2 + \min(5) = 5 + 2 + 5 = 12$$

(2 points)

Step 4: Expand D node, we get:

$$f(F) = g(F) + h(F) = f(D) + 3 = 7 + 3 = 10 \text{ compare it with step 2, then}$$

$$f(F) = 6 \text{ (Destination)} \quad \text{(1 points)}$$

In the end, we find the shortest path from Node S to T using the A* algorithm like: S - A(V1) - T with path 6. (1 points)

